

Teacher Print Queue — Super-Admin First-Time Setup & Governance Guide

Version: v2 (project files: index.html + app.js)

Purpose (for Super-Admin)

This guide provides a first-time setup checklist with step-by-step instructions, recommended governance rules, and an explanation of the built-in reporting/analytics tools so the Super-Admin can run the print queue consistently and auditable.

What the System Is (high level)

- Single-page browser app with Teacher View, Admin View, and Super-Admin controls.
- Primary storage is the browser's localStorage (jobs, settings, notes, logs, grade labels, admin access map).
- Admin login reads credentials from a local server file (admin.env).
- Optional Microsoft sign-in (MSAL) enables direct email sending via Microsoft Graph; otherwise "mailto:" is used.
- Cross-tab "live" refresh relies on the browser storage event (open in more than one tab for sync).

First-Time Setup Checklist (Step-by-Step)

1) Prepare the hosting folder and required files

1. Create a dedicated folder for the app on a local web server or intranet host (e.g., /printqueue/).
2. Place index.html and app.js in the same folder.
3. Ensure the referenced vendor files are present: bootstrap.min.css, bootstrap.bundle.min.js, jquery-3.5.1.slim.min.js, chart.js, vendor/msal-browser.min.js, styles.css (optional).
4. Open index.html through the web server (recommended) rather than file:// to avoid browser restrictions on fetch().
5. Verify the page loads and you can see the Admin View (default) and the "👤 My Jobs" toggle button in the header.

Notes:

- Admin authentication and some settings load via fetch() from the same origin (e.g., ./admin.env and ./msal-config.json).

2) Create admin.env (Admin & Super-Admin accounts)

6. In the same folder as index.html, create a text file named admin.env.
7. Add one account per line using this format: email;password;role
8. Use role = super-admin for the Super-Admin account; use role = admin for standard admins/operators.
9. Save the file and reload the app page.
10. Test login: enter the Super-Admin email and password in “Admin Login” and click Login.

Notes:

- Use strong passwords and restrict file access on the server; this file is fetched by the browser at runtime.
- Keep a break-glass Super-Admin account stored securely (sealed envelope / password vault).

3) Import Teachers (and optionally teacher email addresses)

11. Prepare a .txt file with one teacher per line.
12. For name-only entries, use: Teacher Name
13. To include an email, use: Teacher Name; teacher@schooldomain
14. In Settings & Management, use “Import Teachers & Emails (.txt)” to upload the file.
15. Confirm the Teacher dropdown in “New Print Request” is populated and enabled.

Notes:

- Teacher View name matching is exact; standardise names (e.g., “Surname, Firstname” or “Firstname Surname”) and communicate the convention.
- Email notifications require teacher emails to be stored; importing with emails is recommended.

4) Configure Grade Labels (Super-Admin)

16. Login as Super-Admin.
17. Scroll to the “🔒 Super-Admin” section in Settings & Management.
18. Under “Custom Grade Labels”, enter one label per line (e.g., Grade 8, Grade 9, ...).
19. Click “Save Grade Labels”.
20. Verify the Grade checklist in “New Print Request” shows your labels.

Notes:

- Duplicates and blank lines are ignored; labels become the system’s canonical set for filtering and access control.

5) Assign Admin Grade Access (Super-Admin governance control)

21. Login as Super-Admin.
22. In the “Admin Grade Access” panel, select an admin user.

23. Tick the grade labels this admin is allowed to view/manage.
24. Click "Save Access".
25. Repeat for each admin.
26. Use "Current Assignments" to review the saved access map.

Notes:

- Empty selection means "see all grades" by default; use explicit selections if you want strict separation of duties.

6) Set School Open Hours (Super-Admin)

27. Login as Super-Admin.
28. Click "🕒 Set School Open Hours".
29. For each day, set Enabled/Disabled and set Open/Close times.
30. Click "Save Hours".
31. Test: create a job with a due time outside the configured hours; confirm the system displays an outside-hours warning.

Notes:

- Open hours influence urgency calculations and "short notice" thresholds for more realistic operational planning.

7) Calibrate Time & Queue Behaviour

32. Login as Admin or Super-Admin.
33. Click "⚙️ Adjust Time & Queue Calibration".
34. Set Seconds Per Page, Base Load Time, Quality Check Time, Stapling Time, Trimming Time to match your print room reality.
35. Set Min Lead Time (hrs) for short-notice flags (e.g., 2 hours).
36. Set Max Concurrent Jobs (capacity limit) for capacity warnings (e.g., 2 operators/printers).
37. Set Stall Multiplier (e.g., 1.5) to flag "In Process" jobs that run long.
38. Optional: set Daily Page Cap per Teacher (0 disables).
39. Optional: enable "Require authoriser on every submission".
40. Close the modal to save, then submit a test job and confirm estimates look sensible.

Notes:

- These settings directly affect estimated time, warnings, and analytics "load vs capacity" views.

8) Configure Completion Emails (MailTo vs Microsoft Graph)

41. Login as Admin or Super-Admin.
42. In Settings & Management, enable "Enable mailto: completion emails" if you want completion notifications.

43. Choose a Completed Email Delivery Mode: “MailTo” or “Graph/API”.
44. If using MailTo: test by completing a job and confirm your email client opens with a populated message.
45. If using Graph/API: click “ Microsoft Email Settings”, fill Tenant ID, Client ID, Redirect URI, and Scopes (typically Mail.Send).
46. Use “Import JSON” or “Reload from msal-config.json” if you maintain settings in a file.
47. Click “ Microsoft Sign-In” in the header and sign in as the sending operator account.
48. Complete a test job and confirm the email is sent without opening an email client.

Notes:

- Graph mode requires a properly configured Entra ID (Azure AD) SPA registration and redirect URI alignment.
- The sender mailbox is configured in settings; ensure the mailbox exists and permissions are appropriate.
- If Graph fails, operations may fall back to MailTo depending on the error path.

9) Configure the “Completed Job” Email Template

49. Login as Admin or Super-Admin.
50. Click “ Edit “Completed Job” Email Template”.
51. Edit Subject, Text Body (MailTo), and HTML Body (Graph/API).
52. Use placeholders such as {{reference}}, {{teacher}}, {{grades}}, {{pages}}, {{copies}}, {{due}}, {{completed}}, {{notes}}.
53. Use the preview panel to confirm rendering.
54. Click “Save Template”.

Notes:

- Keep subject lines short; include the job reference for easy searching.

10) Define the standard operating process (SOP) and train staff

55. Decide who is allowed to submit jobs (operators only vs self-service by teachers).
56. Train operators on: submitting jobs, using confirmation modal, marking jobs In Process/Completed, deletion reasons, and re-queueing.
57. Train teachers on Teacher View name matching, how to find jobs, and how to download CSV/notes.
58. Establish a daily routine: review overdue, start queued jobs, close out completed jobs, and export backups.
59. Publish a one-page “How to submit a print job” guide aligned to your school policy.

Recommended Governance Rules (Policy & Controls)

A) Role & Access Governance

- Super-Admin accounts: keep to 1–2 trusted staff; use strong unique passwords; store credentials in a secure vault.
- Admin accounts: create separate accounts per operator (no shared passwords) so audit logs attribute actions correctly.
- Grade access mapping: default should be “see all grades” only if you trust all admins equally; otherwise assign grade labels explicitly.
- Separation of duties: if possible, split “submission” and “completion” responsibilities during high-stakes exam periods.

B) Data & Retention Governance

- Backups: export To-Do and Completed files at least weekly (daily during exam/assessment peaks).
- Retention: define how long completed history should be kept; periodically archive by exporting Completed and clearing old records if required by storage constraints.
- Notes: treat job notes as operational records; export teacher notes when handling disputes or compliance needs.
- Device policy: because localStorage is per-browser, keep operations on a designated “print room workstation” and avoid switching browsers/devices without exporting/importing data.

C) Operational Governance

- Due dates: require “Needed By” for time-sensitive work; discourage ASAP unless truly flexible.
- Urgent flag: allow only when due is within the urgency threshold; review urgent usage weekly for abuse.
- Daily page cap: enable during high-volume terms to control budget and machine wear; define an escalation path for exceptions.
- Deletion reasons: always require a reason; review deletion log monthly to identify recurring issues (duplicate submissions, training gaps).

D) Email & Communications Governance

- Use Graph mode where possible for reliability and consistent formatting; use MailTo as fallback if Graph is unavailable.
- Use a shared mailbox (or no-reply mailbox) for sending completion emails to avoid dependence on one person’s account.
- Template control: limit template editing to Super-Admin; document changes (what/why/date) in a simple change log.
- Privacy: avoid sensitive student information in job notes; keep notes to operational instructions only.

Reporting & Analytics System (What's Available and When to Use It)

1) Admin Dashboard Charts (Quick Operational Snapshot)

What it shows:

- Top 10 teachers by printed volume (pages × copies).
- Completions by hour, split into “on time” vs “late”.

When to use:

- Daily/weekly management check-ins to understand workload distribution and peak completion times.
- To identify teachers/grades driving volume and plan resourcing.

2) Weekly Calendar View

What it shows:

- Completed jobs grouped by day of week, with counts and an estimated total time per day.

When to use:

- End-of-week review to spot days with overload and to compare staffing to workload.
- To communicate expected busy days to leadership or timetable coordinators.

3) Gantt Timeline View

What it shows:

- A time-based bar view for up to ~50 jobs showing request-to-completion durations; late jobs are visually distinct.

When to use:

- Operational diagnostics: quickly see long-running jobs and overall throughput patterns.
- Post-incident analysis when there were missed deadlines or printer downtime.

4) Grade Analytics (Modal) — Detailed Report Suite

Access via “ Grade Analytics”. Filters include range presets (week/month/term/all) and grade multi-select.

Workload

- Jobs per Grade: How many completed jobs each grade generated.
- Est. Time per Grade: Total estimated processing time per grade.
- Print Volume per Grade: Total printed sheets per grade (pages × copies).

Performance

- On-Time vs Late: Counts of on-time vs late completions per grade.
- Turnaround by Grade: Median turnaround (request → completion) per grade.

Trends & Patterns

- Jobs per Grade — Trend: Weekly trend lines per grade (completed jobs).
- On-Time % — Trend: Weekly on-time percentage trend.
- Day × Hour Pattern: Stacked distribution of completions by day and hour.

Bottleneck

- Queue Depth Over Time: Hourly queue depth based on request/completion events.
- Avg Wait by Hour: Average wait (request → started) by submission hour.
- Grade × Day Heatmap: Submission volume by grade vs weekday (Mon–Fri).
- Load vs Capacity: Submitted estimated load compared to capacity line.
- Concurrent In-Process: How many jobs were simultaneously “In process” over time.

Estimation

- Accuracy Distribution: Distribution of actual/estimate ratios for jobs with start times.
- Error by Job Size: Mean estimate error grouped by job size (sheets).
- Error by Task: Mean estimate error grouped by finishing task type.
- Lifecycle Funnel: Average wait, print, total turnaround, and on-time % summary.
- On-Time % + Moving Avg: On-time % trend with moving average smoothing.
- Due Date Slip: How late late jobs were (bucketed).
- Calibration Panel: Suggested time-per-page calibration based on historical jobs with start times.

When to use (examples):

- Budgeting: use “Print Volume per Grade” to estimate paper/toner consumption by grade.
- Service levels: use “On-Time vs Late” and “Due Date Slip” to monitor deadlines.
- Scheduling: use “Queue Depth Over Time” and “Avg Wait by Hour” to align staffing and printer availability to peak demand.
- Process improvement: use “Estimate Accuracy”, “Calibration Panel”, and “Error by Task” to tune calibration settings and reduce bottlenecks.

5) Audit Log (Accountability & Investigations)

What it shows:

- A chronological record of key actions (job created, status changes, imports skipped due to duplicates, deletions) including actor and timestamp.

When to use:

- Dispute resolution: determine who changed a job’s status and when.

- Compliance: provide evidence of operational handling during assessments/exams.
- Training: identify repeated mistakes (e.g., frequent deletions for “submitted in error”).

6) Deletion Log (Reasons & Trend Monitoring)

What it shows:

- A record of deleted jobs capturing reference, teacher, volume, grades, deletion reason, timestamp, and actor.

When to use:

- Monthly governance review to assess whether jobs are being deleted appropriately.
- To identify systemic issues such as duplicate submissions or unclear submission guidance.

7) Import/Export Files (Operational Continuity)

What it does:

- Export To-Do: saves all active (non-completed) jobs to a .txt file for backup or device migration.
- Export Completed: saves completed job history to a .txt file for archiving or analytics continuity.
- Import To-Do / Import History: restore jobs from exported .txt files; duplicates are skipped on history import.

When to use:

- Before switching computers/browsers or clearing cache.
- As a regular backup routine.
- Before/after major policy changes (e.g., new grade labels, new term start).

8) Teacher View Downloads (Teacher Self-Service)

What it provides:

- CSV export of the teacher’s visible jobs (queued, in process, completed).
- Notes export as a plain text file for all teacher job notes.

When to use:

- Teachers tracking their own workload or preparing evidence for class planning.
- When a teacher needs a record of instructions/notes for a job.

Recommended Review Cadence (Super-Admin)

- Daily: check overdue jobs filter; review capacity banner; scan stalled jobs; ensure completion emails are functioning.

- Weekly: review Weekly Calendar + on-time/late patterns; export backups; review audit log sampling.
- Monthly: review deletion log; assess grade access assignments; tune calibration using the Calibration Panel; update teacher list as needed.
- Termly: archive completed history; review policy on urgent flag and daily caps; update grade labels if the curriculum structure changed.

Appendix: File Formats & Conventions

Teacher import file (.txt)

Examples:

Jane Doe

John Smith; john.smith@school.org

admin.env file

Examples:

superadmin@school.org;StrongPasswordHere;super-admin
operator1@school.org;AnotherPassword;admin